

Does OpenAI's Whisper transcribe some English accents less accurately, and does a bigger model close the gap?

Jeo Metelitsa · Project report, October 2026 · Data: Mozilla Common Voice English 27.0 · Code: github.com/jeothecreator/accent-bias-asr

Summary. Whisper makes two to three times as many errors on South Asian, Southern African, Southeast Asian, and second-language European English as it does on US, English, Canadian, and Australian/New Zealand English, and this holds for every model size tested. Larger models reduce errors for every group: the largest model makes about 75% fewer errors than the smallest. But the improvement is smaller in relative terms for the groups that start worse, so the relative gap grows with model size, from 2.1 times to 2.7 times. The difference between groups shrinks in absolute terms (from 15.7 to 4.9 percentage points) while the ratio between them increases, and the increase is statistically reliable.

Speech recognition is a powerful tool used by many around the world. OpenAI claimed that their new automatic speech recognition (ASR) system was to be 75% more accurate, and I wanted to perform some tests to see how it performs under circumstances of different accents, as well as how it copes with different accents saying English by using Mozilla's Common Voice, which is a public dataset of volunteers reading sentences aloud and optionally describing their own accent. I used the English validated set from release 27.0 (September 2026), downloaded from Mozilla Data Collective. Only the metadata and the 3,200 selected audio clips were extracted from the 88 GB download, and no audio is redistributed in the project repository.

Before this test, I read about Braj Kachru and how he categorized language groups. There are three groups: **inner**, **outer**, and **expanding**. Inner groups frequently use the language in their day-to-day life, whereas outer groups use the language, but it is not used abundantly; it is used in government. The expanding languages are languages that are mainly learned as a foreign language.

Hypothesis:

My personal hypothesis is that as the models get larger, for example, from small to large, the model's error rate across all accents will decrease, and I think Canada and the US will perform the best, while expanding accents will perform the worst.

Chart of all the languages and their categorization:

Group	Circle	What it includes	Speakers in sample
US	Inner	United States English, including regional descriptions such as Midwestern	400
England	Inner	English, including regions such as Yorkshire or Liverpool	400
Canada	Inner	Canadian English, including provinces	400
Australia/NZ	Inner	Australian and New Zealand English, pooled	400
South Asia	Outer	India, Pakistan, Sri Lanka, Bangladesh, Nepal	400
Southern Africa	Outer	South Africa, Zimbabwe, Namibia and neighbours	353
SE Asia (PH/MY/SG)	Outer	Philippines, Malaysia, Singapore	347
L2 (European L1)	Expanding	Non-native speakers whose first language is European, such as German, French, or Spanish	238

The first pass allows one clip per speaker, the second allows two, and so on up to a cap of ten. A group only returns to a speaker once it has used every available new speaker. In the end, no speaker contributed more than two clips. Shared sentences second. Within each pass, sentences read by the most groups are taken first, so different groups read the same sentences where possible. 489 sentences appear in at least two groups.

Chart of Models I used:

Model	Parameters	Notes
Whisper tiny	39M	Smallest Whisper model
Whisper base	74M	
Whisper small	244M	
Whisper medium	769M	
Whisper large-v3	1,550M	Largest and most accurate Whisper model
Whisper turbo	809M	A pruned large-v3 with fewer decoder layers, built for speed
wav2vec 2.0 (large, 960h)	317M	Non-Whisper comparison model from Meta, trained only on LibriSpeech audiobooks

I made sure to keep all the models controlled, so these things were kept in mind:

- All languages were fixed to English
- All beam search with 5 beams and temperature fallback
- All ran on an online t4 gpu from Google Colab, which ran for 4 hours total

How to catch errors:

Errors were able to be caught because before scoring, both the reference sentence and the model's transcript go through Whisper's English text normalizer, so differences in punctuation, capitalization, spelling variety, or number format ("Mr." and "mister", "colour" and "color") are not counted as errors. It then goes through the formula:

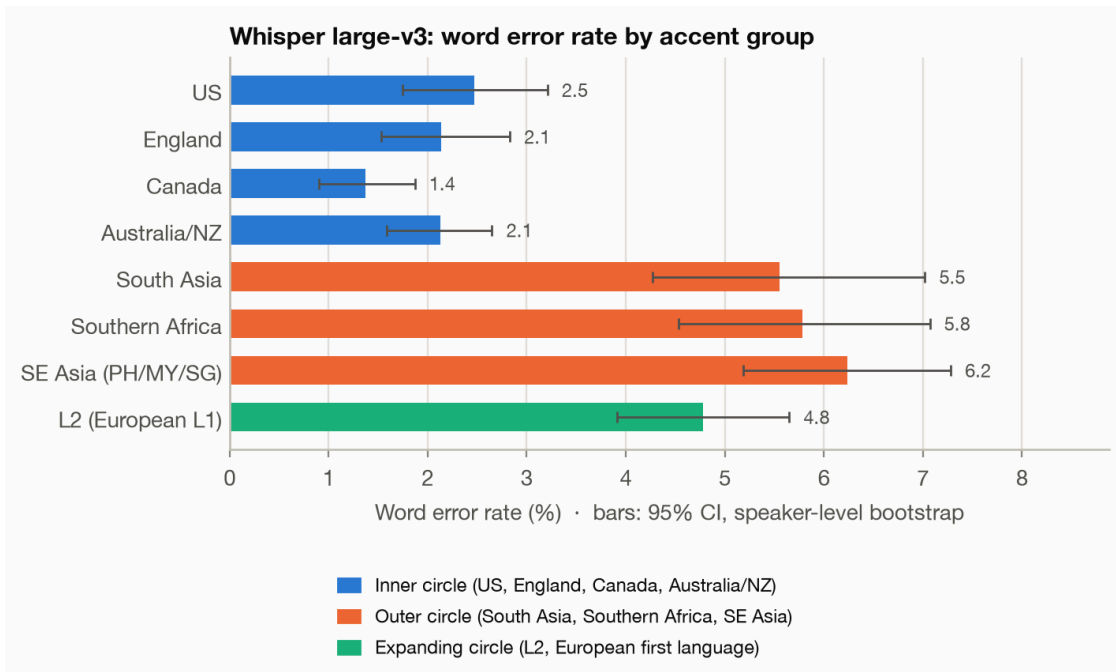
$$WER = (substitutions + deletions + insertions) / words\ in\ the\ reference$$

For example: "what joy there is in living" → what joy there is living" would be one error out of six, so ⅙ or 16.7% error rate.

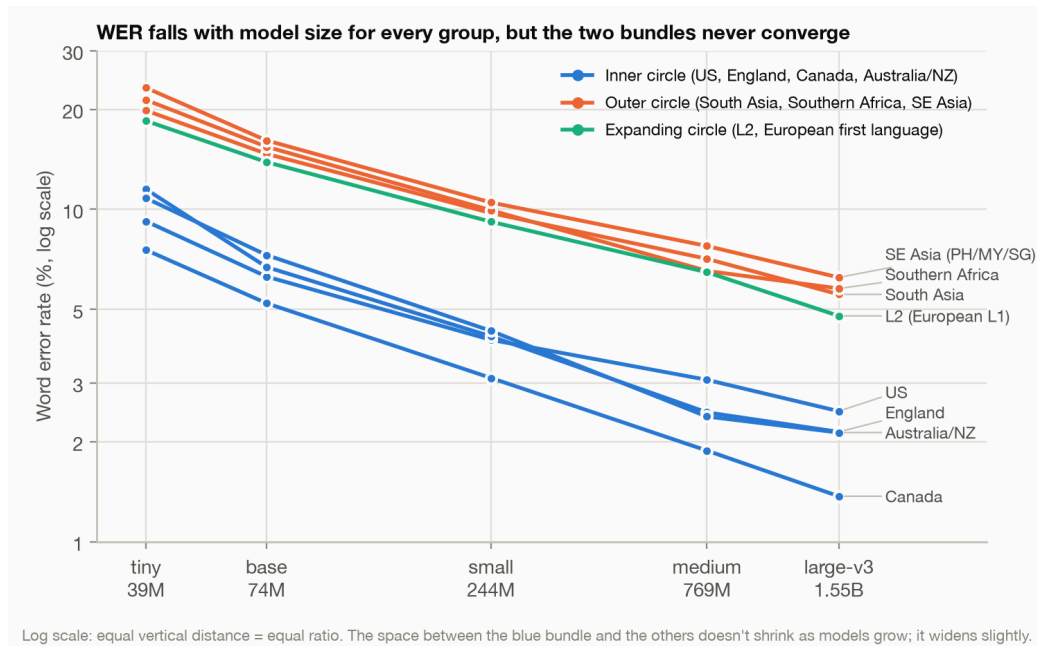
Results Table:

Group	tiny	base	small	medium	large-v3	turbo	wav2vec2
US	9.2	6.3	4.1	3.1	2.5	2.4	5.2
England	11.5	6.7	4.2	2.5	2.1	2.2	6.1
Canada	7.6	5.2	3.1	1.9	1.4	1.5	6.0
Australia/N Z	10.8	7.3	4.3	2.4	2.1	2.6	7.3
South Asia	19.8	14.7	9.7	7.1	5.6	6.8	14.7
Southern Africa	21.4	15.4	9.9	6.5	5.8	7.5	14.7
SE Asia (PH/MY/S G)	23.2	16.1	10.5	7.8	6.2	7.1	17.1
L2 (European L1)	18.5	13.9	9.2	6.5	4.8	5.5	13.8

Results (Graph):



*This chart represents the final results, where it is clear to see how, on the large model (the most accurate model), Canada, England, and Australia perform the best on Whisper, whereas Southeast Asia as well as Southern Africa have the highest error rate when transcribing.



*This graph signifies how the error rate for all accents went down significantly depending on the parameters of the model, with large having the lowest rate compared to all the other models, with a peak of only a 1.4% error rate, which is almost close to perfect.

Further Observations:

I also noticed that the Turbo model, which is designed to be a faster large v3, was about as accurate for the inner circle, but way less accurate for the other groups.

Where the models failed

The 50 clips with the highest large-v3 WER include several hallucinations, where Whisper wrote fluent text that has nothing to do with the audio, such as "Transcription by CastingWords" or "Hello friends. Today, I will explain to you...". These are a known failure of Whisper, and a single one can add many insertion errors.

To check that these clips are not driving the result, I removed the 82 clips on which any model scored a WER of 100% or more and recalculated the ratio. The ratio is a little lower without them, but the pattern is the same: it still rises from 1.95 (tiny) to 2.47 (large-v3). The removed clips come mostly from South Asia (21) and SE Asia (20), so whether they reflect the model failing on those accents or problems with the recordings is an open question, addressed in the next steps below.

Model	Ratio, all clips	Ratio without the 82 clips
tiny	2.11	1.95
base	2.36	2.18
small	2.51	2.34
medium	2.84	2.64
large-v3	2.74	2.47
turbo	3.06	2.76
wav2vec2	2.46	2.31

What these results mean:

Based on the results, my hypotheses were supported, and the result is more noticeable than I predicted. A speaker from Asia or Africa is almost twice as likely to get an error compared to the other accent groups. OpenAI's claim about how the new model "cuts errors by 75%" - which is true on average - but the benefit is not shared evenly across all accents of English.

What did I learn from this project:

I felt like I learned so many things from this project. Firstly, I learned how to use Google Colab to run cells and run scripts on a virtual t4 GPU. Secondly, I learned how to download datasets such as the Mozilla English Dataset and how to apply them to my project. Lastly, I learned how to document all my research and how to use matplotlib to graph all my data into visually aesthetic charts.

Some Ethics of this project

- Results are reported only for groups, never for individuals. Common Voice speaker IDs were replaced with project-only IDs, and no attempt was made to identify anyone.
- Common Voice audio is not redistributed. The repository contains clip IDs, code and processing steps, so others can download the data and reproduce the work themselves.
- Findings describe where the system underperforms, not how anyone should speak.

How to reproduce this

All code is at github.com/jeothecreator/accent-bias-asr, with automated tests and a step-by-step README.

Phase	What it does	Command
2. Data	Maps accent labels, filters and samples clips, writes the data card	<pre>python src/prepare_dataset.py --cv-dir <corpus>/en</pre>
3. Transcribe	Runs each model in Colab and saves transcripts as it goes	<pre>notebooks/03_transcribe.ipynb</pre>
3. Score	Normalizes text and computes WER per clip	<pre>python src/score.py</pre>
4. Analyze	Bootstrap intervals, gap statistics, charts, and worst-clips list	<pre>python src/analyze.py</pre>

The pipeline code was written with the help of an AI coding assistant (Claude). The research question, the decisions recorded in the data card, the Colab runs, and the interpretation are part of this project's documented work.

Appendix: results for every model

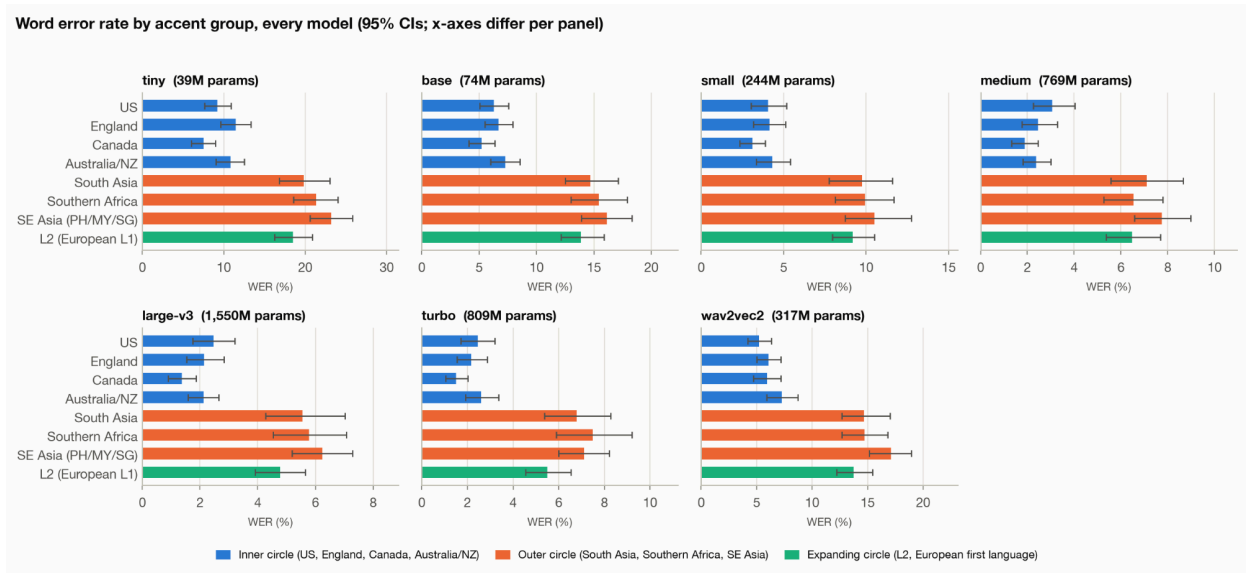


Figure A1. WER by accent group for all seven models, with 95% confidence intervals.

References

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